

Hackathon 1

Oves Badami

August 21, 2026

Problem 1

Consider yourself an engineer assigned to fix the patriot Missile Problem.

Background: In the case of a Patriot Missile accident, a radar would scan the atmosphere for an incoming missile and track its trajectory. The scanning rate is limited to 0.1 sec steps, and hence the radar keeps time records in 0.1 sec intervals. The scanned data is then given to the High Performance Computing Facility, which processes it with regard to the tracking of the incoming missile trajectory. The limited radar's precision led to incorrect time storage and trajectory calculations.

Problem Statement: Can you write simple code using float variables that can track the time in steps of 0.1 sec (with float precision). The maximum error ($\text{fabs}(\text{actual time} - \text{calculated time})$) in the time calculation that can be accumulated over the entire operation range (10 hours) has to be less than 10 ms. Write a C code that satisfies the above constraint of error accumulation and show the error accumulated over a period of 10 hours of operation (i.e. 36000 secs).

There will be a limitation to the maximum time that you can represent because of the overall float data type for time, but that is much larger than 10 hours.

Problem 2

Calculate the sum of the first 100000 terms of the harmonic series using the float data type.